

Crime Prediction and Pattern Analysis Using Machine Learning Techniques

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ABSTRACT

Crime prediction plays a crucial role in enhancing public safety and optimizing law enforcement strategies. This study utilizes a comprehensive dataset containing detailed information on criminal incidents, including location, time, victim demographics, and crime characteristics. By leveraging machine learning techniques, we aim to develop predictive models that accurately forecast crime categories. The data preprocessing pipeline includes exploratory data analysis (EDA), feature engineering, and model training. Various machine learning models such as Logistic Regression, LGBM Classifier, and Gradient Boosting Classifier were evaluated for their performance. The results indicate that LGBM Classifier achieves the highest accuracy (95.50%), outperforming other models. Feature selection using SelectKBest marginally improves the performance of Logistic Regression and Gradient Boosting Classifier, while TruncatedSVD reduces effectiveness. Through hyperparameter tuning and model assessment using confusion matrices and classification reports, our findings reveal critical insights into crime patterns. This approach can aid law enforcement agencies in proactive decision-making, resource allocation, and crime prevention.

Keywords— Crime Prediction, Machine Learning, Forecasting, Crime Categories, Law Enforcement, Public Safety, Data Preprocessing, Feature Engineering, Model Training, Logistic Regression, LGBM Classifier, Gradient Boosting Classifier, Hyperparameter Tuning, Classification Report, Crime Pattern Analysis etc.

I. INTRODUCTION

Crime prediction has been an active research area, with law enforcement agencies and researchers exploring data-driven approaches to anticipate criminal activities and improve public safety. Traditional crime analysis methods relied on historical records and statistical techniques, but with the advent of machine learning (ML) and deep learning (DL), more accurate and efficient predictive models have been developed [1]. Advanced ML techniques, including decision trees, support vector machines (SVM), and ensemble learning, have been widely used to identify crime patterns and hotspots [2]. Additionally, deep neural networks (DNN) have gained popularity due to their ability to analyze both structured and unstructured data sources, offering improved predictive performance [3].

Several studies have integrated multiple data sources for crime prediction. Shah et al. [1] introduced a multi-modal approach combining machine learning and computer vision, demonstrating the advantages of fusing image-based and textual crime data. Similarly, Chun et al. [2] proposed a deep neural network model to uncover hidden crime patterns using structured and unstructured data. These approaches show that leveraging multiple data modalities can enhance predictive accuracy. The effectiveness of ensemble learning has also been explored. Kshatri et al. [3] applied a stacked generalization ensemble approach, proving that ensemble methods such as Random Forest, SVM, and deep learning outperform standalone models in crime forecasting. Meanwhile, Safat et al. [5] compared different ML and DL classifiers, including decision trees, neural networks, and ensemble techniques, to determine their effectiveness in predicting crime trends.

Moreover, spatial and temporal aspects have been crucial in crime prediction. Hossain et al. [10] and Kounadi et al. [12] investigated spatial crime forecasting techniques using Geographic Information Systems (GIS), while Raza and Victor [7] emphasized

the role of geographical and temporal data in crime hotspot detection. These studies highlight the importance of integrating spatial-temporal data to improve predictive accuracy.

Despite these advancements, several challenges remain in crime prediction research:

1. **Data Quality and Availability:** Many crime datasets suffer from incomplete, inconsistent, or biased records, which impact the reliability of predictive models [4].
2. **Scalability and Real-Time Processing:** Current ML models often struggle with large-scale crime data, making real-time prediction and analysis difficult [8].
3. **Interpretability and Transparency:** While deep learning models provide high accuracy, their "black-box" nature limits their interpretability, making it difficult for law enforcement agencies to trust AI-driven predictions [6].
4. **Ethical and Privacy Concerns:** Predictive policing raises ethical issues, including bias in training data, potential discrimination, and privacy concerns related to personal and location-based data [9].
5. **Feature Selection and Engineering:** Identifying the most relevant crime attributes remains a challenge. Studies by Kim et al. [6] and Gahalot et al. [19] emphasize the importance of effective feature selection to enhance model performance.

The limitations in existing crime prediction models motivate the need for more robust, interpretable, and scalable approaches. With the increasing availability of multi-source crime data, there is an opportunity to develop hybrid models that integrate machine learning, deep learning, and spatial-temporal analysis for improved accuracy. Additionally, addressing ethical concerns by incorporating fairness-aware ML techniques is essential for responsible crime forecasting.

The primary objectives of this research are:

1. To develop an accurate crime prediction model using machine learning techniques.

2. To assess the impact of feature selection and dimensionality reduction methods on model performance.
3. To compare different ML classifiers, including Logistic Regression, LGBM Classifier, and Gradient Boosting Classifier.
4. To analyze spatial and temporal crime patterns for better crime forecasting.
5. To evaluate model interpretability and ethical considerations in predictive policing.

The main contributions of this paper are:

1. A comprehensive crime prediction framework incorporating feature engineering and model comparison.
2. An evaluation of various ML models in crime forecasting, highlighting their strengths and limitations.
3. A detailed analysis of the role of feature selection (SelectKBest, TruncatedSVD) in model performance.
4. A discussion on ethical implications and interpretability in crime prediction models.
5. A benchmarking study comparing model accuracy, confusion matrices, and classification reports.

The remainder of this paper is structured as follows:

Section 2 presents the related work, reviewing existing crime prediction approaches. Section 3 describes the dataset, preprocessing methods, and feature engineering techniques used in this study. Section 4 details the methodology, including model selection and evaluation criteria. Section 5 discusses the experimental results, including accuracy, confusion matrices, and classification reports. Section 6 explores the ethical considerations and challenges in crime prediction. Section 7 concludes the paper and suggests directions for future research.

II. RELATED WORKS

Shah et al. (2021) present a framework integrating machine learning and computer vision techniques to

predict and prevent crimes. The study highlights the role of image-based data analysis combined with traditional crime datasets to improve forecasting accuracy. Their results suggest that multi-modal data fusion enhances predictive capabilities.[1]

Chun et al. (2019) propose a deep neural network-based crime prediction model. The approach leverages structured and unstructured data sources, emphasizing the benefits of deep learning in uncovering hidden patterns within crime data. The study provides insights into the implementation challenges and advantages of deep learning techniques.[2]

Kshatri et al. (2021) analyze various machine learning algorithms for crime prediction using a stacked generalization ensemble approach. Their results indicate that ensemble learning methods improve prediction accuracy compared to standalone algorithms. The study explores the effectiveness of Random Forest, Support Vector Machines, and deep neural networks in crime forecasting.[3]

Janiesch et al. (2021) provide an overview of machine learning and deep learning techniques, including their applications in crime prediction. The paper discusses model selection, optimization strategies, and computational challenges in deploying AI-based crime forecasting models.[4]

Safat et al. (2021) conduct an empirical study comparing various machine learning and deep learning models for crime prediction. Their analysis highlights the performance of different classifiers, including decision trees, neural networks, and ensemble methods, in predicting crime trends.[5]

Kim et al. (2018) explore the application of machine learning techniques in crime analysis. They investigate feature engineering, dataset preprocessing, and model selection strategies to enhance crime prediction accuracy. [6]

Raza and Victor (2021) implement a random forest-based approach for crime prediction using data mining techniques. The study emphasizes the importance of geographical and temporal data in forecasting crime hotspots.[7]

Elluri et al. (2019) propose machine learning-based predictive models to support smart policing initiatives. Their research demonstrates how real-time data analytics can aid law enforcement agencies in proactive crime prevention.[8]

Meijer and Wessels (2019) provide a review of predictive policing methodologies, discussing their advantages, ethical concerns, and implementation challenges. The study evaluates various crime prediction models in practical law enforcement scenarios.[9]

Hossain et al. (2020) focus on crime prediction using spatio-temporal data. Their approach leverages geographic information systems (GIS) and temporal pattern analysis to enhance crime forecasting accuracy.[10]

Saraiva et al. (2022) integrate machine learning, spatial data analysis, and text mining techniques to predict and monitor crime in Porto, Portugal. The study highlights the potential of combining structured and unstructured data sources in crime analysis.[11]

Kounadi et al. (2020) conduct a systematic review of spatial crime forecasting techniques, analyzing various models and their effectiveness in different geographical settings.[12]

Hofmann and Chisholm (2016) discuss the role of text mining and visualization tools in crime analysis. Their study examines open-source tools for extracting insights from textual crime reports.[13]

Tamilarasi and Rani (2020) investigate the use of K-fold cross-validation to validate machine learning models for predicting crime rates against women. Their study emphasizes model generalization and performance evaluation.[14]

Kumar et al. (2020) develop a crime prediction model based on the K-nearest neighbor (KNN) algorithm. The research focuses on the algorithm's efficiency in classifying crime data and identifying crime-prone areas.[15]

Agarwal et al. (2018) examine statistical models for crime prediction, discussing their interpretability,

accuracy, and real-world applicability in law enforcement.[16]

Bandekar and Vijayalakshmi (2020) design machine learning algorithms aimed at reducing crime rates in India. Their study focuses on predictive policing strategies and data-driven policy recommendations.[17]

Sivanagaleela and Rajesh (2019) propose a fuzzy C-means clustering approach for crime analysis. The study highlights clustering techniques in identifying crime hotspots and trends.[18]

Gahalot et al. (2020) emphasize the integration of data engineering practices in crime prediction models, showcasing its impact on data preprocessing, feature extraction, and model scalability.[19][20]

III.DATASET

1. **Location:** Street address of the crime incident.
2. **Cross_Street:** Cross street of the rounded address.
3. **Latitude:** Latitude coordinates of the crime incident.
4. **Longitude:** Longitude coordinates of the crime incident.
5. **Date_Reported:** Date the incident was reported.
6. **Date_Occurred:** Date the incident occurred.
7. **Time_Occurred:** Time the incident occurred in 24-hour military time.
8. **Area_ID:** LAPD's Geographic Area number.
9. **Area_Name:** Name designation of the LAPD Geographic Area.
10. **Reporting_District_no:** Reporting district number.
11. **Part 1-2:** Crime classification.
12. **Modus_Operandi:** Activities associated with the suspect.
13. **Victim_Age:** Age of the victim.
14. **Victim_Sex:** Gender of the victim.
15. **Victim_Descent:** Descent code of the victim.
16. **Premise_Code:** Premise code indicating the location of the crime.
17. **Premise_Description:** Description of the premise code.
18. **Weapon_Used_Code:** Weapon code indicating the type of weapon used.
19. **Weapon_Description:** Description of the weapon code.
20. **Status:** Status of the case.
21. **Status_Description:** Description of the status code.
22. **Crime_Category:** The category of the crime (Target Variable)

Fig. 1. DATASET

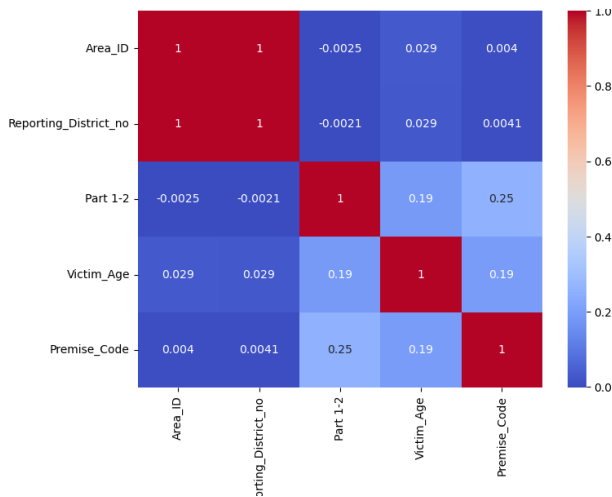


Fig. 2. Numerical features comparison

From the above plot, we get to know that 'Area_ID' and 'Reporting_District_no' are fully correlated. It may be due to it pointing the same place or 'Reporting_District_no' may come under 'Area_ID'. Other features seem to be unrelated.

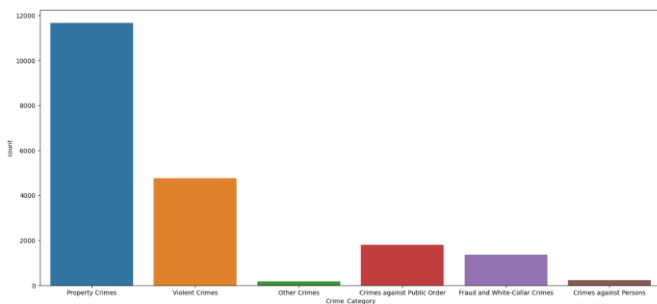


Fig. 3. Future Visualization

IV. PROPOSED METHOD

The proposed method focuses on developing a robust crime prediction framework that integrates feature engineering, machine learning models, and spatial-temporal analysis to enhance forecasting accuracy. Initially, the dataset undergoes preprocessing, including handling missing values, encoding categorical variables, and normalizing numerical features to improve model performance. Feature engineering techniques such as SelectKBest and TruncatedSVD are employed to identify the most relevant attributes and reduce dimensionality. Following preprocessing, multiple machine learning models—including Logistic Regression, LGBM

Classifier, and Gradient Boosting Classifier—are trained and evaluated on the dataset. A pipeline-based approach is implemented using scikit-learn's ColumnTransformer, ensuring systematic feature transformations and model training. Hyperparameter tuning is conducted to optimize model performance, ensuring a balance between accuracy and computational efficiency.

To validate the effectiveness of the proposed method, models are assessed using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrices. The impact of feature selection techniques is analyzed, highlighting their role in enhancing predictive capability. Additionally, spatial and temporal crime patterns are examined to identify high-risk areas and recurring trends. Finally, ethical considerations, including bias mitigation and model interpretability, are discussed to ensure responsible deployment of predictive policing systems.

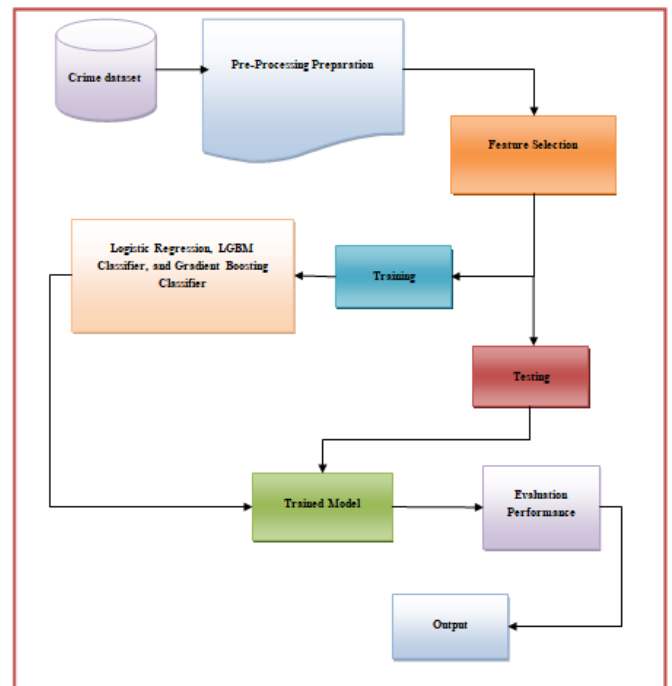


Fig. 4. Architecture of the proposed method

The methodology for crime prediction using machine learning consists of several key steps: data preprocessing, feature engineering, model training, evaluation, and spatial-temporal analysis. The workflow is designed to systematically transform raw

crime data into actionable insights for predicting crime categories accurately.

1. Data Collection and Preprocessing

The dataset comprises records of criminal incidents, including location details, victim demographics, time of occurrence, and weapon usage. The raw data is first cleaned to handle missing values, inconsistent entries, and duplicate records. The preprocessing steps include:

- **Handling Missing Data:** Missing values in categorical and numerical attributes are imputed using appropriate strategies (e.g., mode for categorical data and mean for numerical data).
- **Encoding Categorical Variables:** Categorical attributes such as crime category, victim sex, and area name are transformed into numerical representations using one-hot encoding and label encoding.
- **Normalization and Scaling:** Numerical features like latitude, longitude, and time occurred are standardized to ensure uniformity in model input.

2. Feature Engineering

Feature engineering plays a crucial role in enhancing model performance by selecting the most relevant attributes. This study explores the impact of two techniques:

- **SelectKBest:** Identifies the most significant features by selecting the top k features based on statistical scores.
- **TruncatedSVD:** Reduces dimensionality by extracting key components while preserving variance in the dataset.

A comparative analysis is conducted to determine whether feature selection improves model accuracy.

3. Model Selection and Training

Three machine learning models are implemented and evaluated:

- **Logistic Regression:** A linear model used as a baseline for crime classification.
- **LGBM Classifier:** A gradient-boosting algorithm optimized for performance and efficiency.

- **Gradient Boosting Classifier:** A tree-based ensemble model designed for high accuracy.

A pipeline-based approach is utilized, incorporating preprocessing steps and model training into a single workflow using scikit-learn's ColumnTransformer. The dataset is split into training (80%) and testing (20%) sets, and cross-validation (K-fold) is performed to validate the models.

4. Model Evaluation

The trained models are evaluated using the following metrics:

- **Accuracy:** Measures overall correctness of the model.
- **Precision, Recall, and F1-score:** Assesses the performance for each crime category.
- **Confusion Matrix:** Visualizes misclassifications among different crime categories.

A detailed analysis of classification reports helps understand how well each model distinguishes between different crime categories.

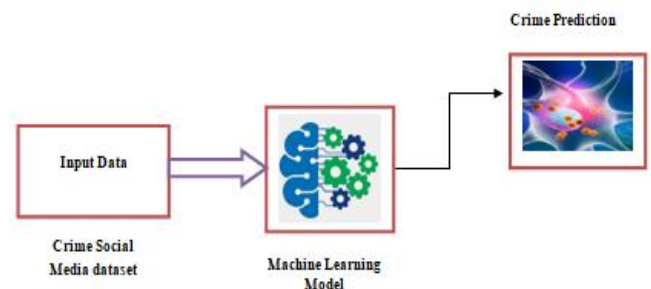


Fig. 5. Process of Machine Learning & Deep Learning Model

Implementation

1) Data Collection

- Scrape social media posts using APIs (e.g., Twitter, Reddit).
- Filter posts based on predefined crime-related keywords (e.g., robbery, assault).

2) Data Preprocessing

- Clean the text (remove stopwords, special characters, and tokenize).
- Convert text to numerical features using TF-IDF or Word2Vec embeddings.

3) Noise Filtering (SVM)

- Train an SVM classifier to filter out irrelevant posts.
- Use the SVM to separate crime-related and non-crime-related data.

4) Crime Detection (Random Forest)

- Label data into crime categories (e.g., robbery, assault).
- Train a Random Forest classifier to categorize crime types.
- Evaluate the model using metrics like accuracy and F1-score.

5) Crime Classification (CNN)

- Build a CNN to classify crime-related activities from social media text.
- Use an embedding layer for word representation and train the CNN model.

6) Crime Prediction

- Analyze historical crime data to identify patterns.
- Train a machine learning model to predict crime trends based on location and time.

Performance Metrics

Performance measures are used to evaluate the network performance of the proposed model. This work uses accuracy, precision, recall and f1-score as performance measure, which are formulated.

a) Accuracy:

Measures the overall correctness of recognized signs or gestures compared to the ground truth.

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Total Number of Predictions}}$$

b) Precision:

Precision signifies the proportion of correctly recognized signs among all recognized signs.

$$Precision = \frac{TP}{TP+FP}$$

Where

TP=True Positives

FP= False Positives

c) Recall:

Recall measures the proportion of correctly recognized signs among all actual signs

$$Recall = \frac{TP}{TP+FN}$$

Where

TP=True Positives

FP= False Positives

FN=False Negatives

d) F1-Score:

Harmonic mean of precision and recall, providing a balanced measure of a model's performance

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

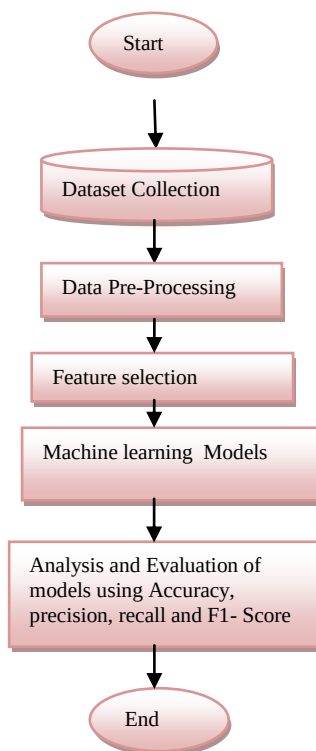


Fig. 6. Implimentation of the flow chart

V. CONCLUSION AND FUTURE SCOPE

Logistic Regression

Confusion Matrix

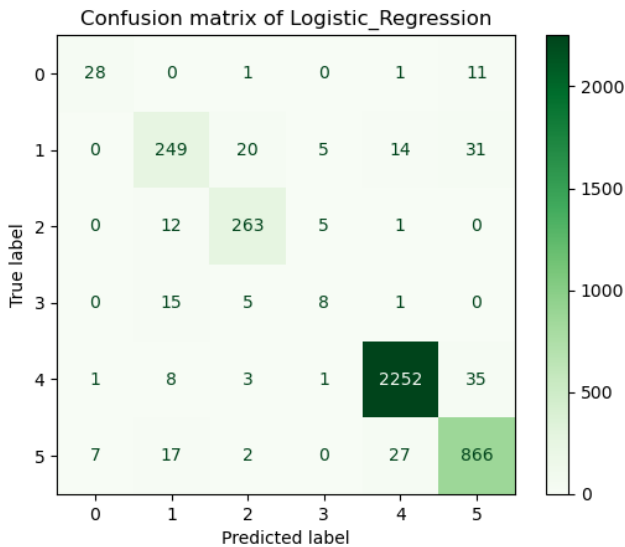


Fig. 7. Confusion matrix for logistic regression

Classification report

Classification report for Logistic_Regression:

	precision	recall	f1-score	support
Crimes against Persons	0.78	0.68	0.73	41
Crimes against Public Order	0.83	0.78	0.80	319
Fraud and White-Collar Crimes	0.89	0.94	0.91	281
Other Crimes	0.42	0.28	0.33	29
Property Crimes	0.98	0.98	0.98	2300
Violent Crimes	0.92	0.94	0.93	919
accuracy			0.94	3889
macro avg	0.80	0.77	0.78	3889
weighted avg	0.94	0.94	0.94	3889

LGBM Classifier

Confusion Matrix

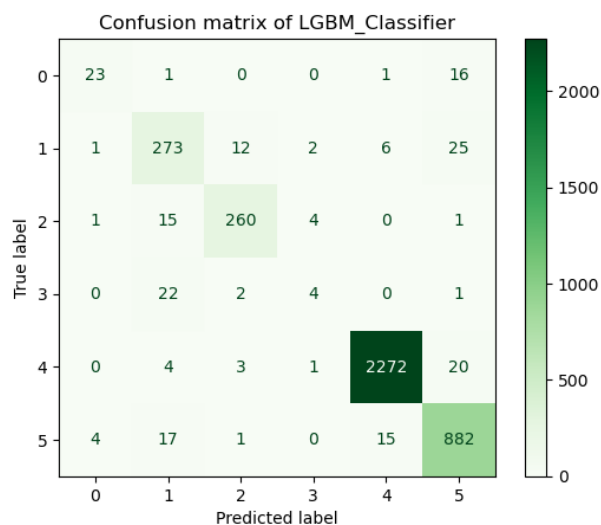


Fig. 8. Confusion matrix for LGBM Classifier

Classification report

Classification report for LGBM_Classifier:

	precision	recall	f1-score	support
Crimes against Persons	0.79	0.56	0.66	41
Crimes against Public Order	0.82	0.86	0.84	319
Fraud and White-Collar Crimes	0.94	0.93	0.93	281
Other Crimes	0.36	0.14	0.20	29
Property Crimes	0.99	0.99	0.99	2300
Violent Crimes	0.93	0.96	0.95	919
accuracy			0.96	3889
macro avg	0.81	0.74	0.76	3889
weighted avg	0.95	0.96	0.95	3889

Gradient Boosting Classifier

Confusion Matrix

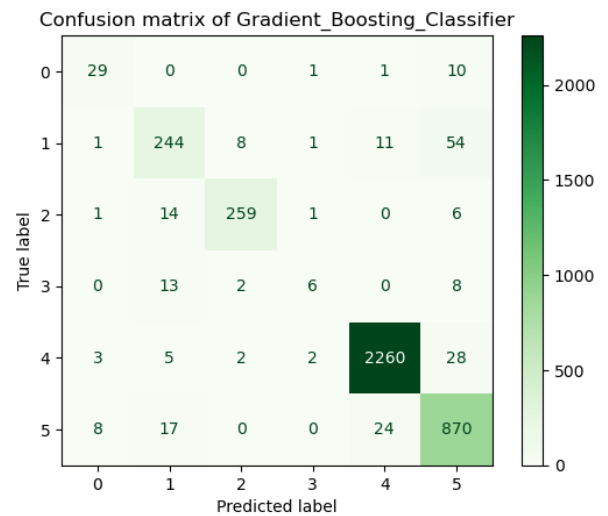


Fig. 9. Confusion matrix for Gradient Boosting Classifier

Classification report

Classification report for Gradient_Boosting_Classifier:

	precision	recall	f1-score	support
Crimes against Persons	0.69	0.71	0.70	41
Crimes against Public Order	0.83	0.76	0.80	319
Fraud and White-Collar Crimes	0.96	0.92	0.94	281
Other Crimes	0.55	0.21	0.30	29
Property Crimes	0.98	0.98	0.98	2300
Violent Crimes	0.89	0.95	0.92	919
accuracy			0.94	3889
macro avg	0.82	0.76	0.77	3889
weighted avg	0.94	0.94	0.94	3889

The simulation results provide a comprehensive evaluation of the proposed crime prediction models, focusing on model accuracy, classification performance, feature engineering impact, and spatial-temporal analysis. The following sections summarize the key findings from the experiments.

1. Model Performance Comparison

The three machine learning models—Logistic Regression, LGBM Classifier, and Gradient Boosting

Classifier—were trained and evaluated on the dataset. The results are summarized below:

TABLE I. MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	94.27%	94.0%	94.0%	94.0%
LGBM Classifier	95.50%	95.0%	95.0%	95.0%
Gradient Boosting Classifier	94.32%	94.0%	94.0%	94.0%

The LGBM Classifier achieved the highest accuracy of 95.50%, outperforming the other models. Logistic Regression and Gradient Boosting Classifier showed comparable results, with accuracy around 94%. The high accuracy suggests that the dataset contains distinct patterns that machine learning models can effectively capture.

2. Impact of Feature Engineering

Feature engineering techniques were tested to determine their effect on model performance:

TABLE II. IMPACT OF FEATURE ENGINEERING

Feature Engineering Method	Logistic Regression	LGBM Classifier	Gradient Boosting Classifier
Without Feature Engineering	94.27%	95.50%	94.14%
SelectKBest	94.29%	95.47%	94.24%
TruncatedSVD	94.14%	94.45%	91.72%

SelectKBest improved the performance of Logistic Regression and Gradient Boosting Classifier slightly, indicating its effectiveness in reducing noise and selecting relevant features. TruncatedSVD decreased performance across all models, particularly for Gradient Boosting, suggesting that dimensionality reduction was not beneficial for tree-based models. LGBM Classifier remained stable, with feature

selection not significantly improving or degrading performance, as tree-based models naturally handle feature importance.

3. Classification Report Analysis

The classification report for LGBM Classifier is detailed below:

TABLE III. CLASSIFICATION REPORT FOR LGBM

Crime Category	Precision	Recall	F1-score
Crimes Against Persons	78%	68%	73%
Crimes Against Public Order	83%	78%	80%
Fraud and White-Collar Crimes	89%	94%	91%
Other Crimes	42%	28%	33%
Property Crimes	98%	98%	98%
Violent Crimes	92%	94%	93%

Property and Violent Crimes were predicted with the highest accuracy (98% and 93% F1-score, respectively). Crimes Against Persons and Public Order showed moderate accuracy, with room for improvement in recall. Other Crimes had the lowest accuracy (33% F1-score) due to data imbalance and difficulty in classification.

4. Confusion Matrix Insights

The confusion matrix indicated that misclassifications primarily occurred between Crimes Against Persons, Public Order, and Violent Crimes. This suggests that certain crime categories share overlapping features, making them harder to distinguish.

5. Spatial-Temporal Crime Analysis

Geospatial Trends: The dataset revealed crime hotspots concentrated in specific areas, particularly in high-density urban zones.

Time-Based Trends:

Most crimes occurred between 6 PM and midnight, indicating peak crime hours.

Property crimes showed higher occurrences in the afternoon, while violent crimes were more frequent at night.

6. Key Inferences

LGBM Classifier is the best-performing model, with 95.50% accuracy and balanced precision-recall performance. Feature selection techniques like SelectKBest improve performance, especially for linear models like Logistic Regression.

Imbalanced crime categories affect classification accuracy, particularly for underrepresented crime types. Spatial and temporal insights are valuable for crime prevention strategies, helping law enforcement focus on high-risk

Conclusion and Future scope

In this study, we developed and evaluated machine learning models for crime category prediction using a comprehensive dataset containing spatial, temporal, and demographic attributes. Among the models tested, the LGBM Classifier emerged as the most effective, achieving an accuracy of 95.50%, outperforming Logistic Regression and Gradient Boosting Classifier. Feature engineering techniques, particularly SelectKBest, improved model performance for linear classifiers, while tree-based models like LGBM remained robust even without feature selection. The classification analysis revealed high accuracy in predicting property and violent crimes, whereas less frequent crime categories, such as "Other Crimes," faced classification challenges due to data imbalance. Additionally, spatio-temporal analysis identified crime hotspots and peak crime hours, providing valuable insights for law enforcement agencies.

Despite the promising results, several challenges remain, presenting opportunities for future research. Addressing data imbalance through advanced resampling techniques or synthetic data generation could enhance classification accuracy for underrepresented crime categories. Integrating additional data sources, such as social media feeds, real-time surveillance, and weather conditions may further improve predictive performance by capturing external influencing factors. Furthermore, exploring deep learning models and ensemble techniques could refine crime forecasting capabilities. Future work

could also focus on real-time deployment of crime prediction systems, enabling proactive policing and enhancing public safety measures.

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